FOR A FLAT LOCAL HOMOMORPHISM THE DETERMINANT OF A BASE CHANGED ENDOMORPHISM IS THE M-TH POWER OF THE ORIGINAL DETERMINANT

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The statement refers to the exponent m defined in its own first display and to the object (M, varphi) 'as above', both delivered verbatim. The three references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- The determinant det_kappa(varphi) of an endomorphism of a finite length module over the residue field, and its multiplicativity along short exact sequences. invoked through the earlier lemma at proof lines 36735 and 36755. Is the invariant being base changed and the tool that reduces to simple objects.
- Flatness of the local homomorphism, giving base change of short exact sequences. used at proof lines 36733-36734. Is what makes the reduction to simple objects legitimate.
- Compatibility of determinants of linear maps with field extensions. used at proof lines 36763-36764. Identifies the determinant over kappa' of the base changed map on the simple quotient with the m-th power of the determinant over kappa.

Source reference. Original lemma 36678: sources/stacks-more-algebra/pinned-original-source.tex, source lines 36678–36678. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let $(R, \mathfrak{m}, \kappa) \to (R', \mathfrak{m}', \kappa')$ be a flat local homomorphism of local rings such that $m = length_{R'}(R'/\mathfrak{m}R') < \infty$. For any (M, φ) as above, the element $\det_\kappa(\varphi)^m$ maps to $\det_{\kappa'}(\varphi \otimes 1 : M \otimes_R R' \to M \otimes_R R')$ in κ'.*

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Proof. The flatness of $R \to R'$ assures us that short exact sequences as in Lemma 36678 base change to short exact sequences over R'. Hence by the multiplicativity of Lemma 36678 we may assume that (M, φ) is a simple object of our category (see introduction to this section). In the simple case M is annihilated by $\mathfrak{m}$. Choose a filtration

$$0 \subset I_1 \subset I_2 \subset \ldots \subset I_{m-1} \subset R'/\mathfrak{m}R'$$

whose successive quotients are isomorphic to κ' as R'-modules. Then we obtain the filtration

$$0 \subset M \otimes_\kappa I_1 \subset M \otimes_\kappa I_2 \subset \ldots \subset M \otimes_\kappa I_{m-1} \subset M \otimes_\kappa R'/\mathfrak{m}R' = M \otimes_R R'$$

whose successive quotients are isomorphic to $M \otimes_\kappa \kappa'$. Also, these submodules are invariant under $\varphi \otimes 1$. By Lemma 36678 we find

$$\det_{\kappa'}(\varphi \otimes 1 : M \otimes_R R' \to M \otimes_R R') = \det_{\kappa'}(\varphi \otimes 1 : M \otimes_\kappa \kappa' \to M \otimes_\kappa \kappa')^m = \det_\kappa(\varphi)^m$$

The last equality holds by the compatibility of determinants of linear maps with field extensions. This proves the lemma. $\square$